Hardware and training setup

Standardised comparison | 1 November 2026

Hardware

CPU: AMD Ryzen 7 9800X3D (4 PyTorch threads used).
GPU: AMD Radeon RX 9070 XT; all models trained on GPU.
Software: Windows, PyTorch 2.12.0+rocm7.14.0; NumPy 2.5.4; h5py 3.16.0.
Precision: FP32. **Data-loader workers:** 4.

Data and training budget

Setting	Shared configuration
Input / SNR range	50 x 100 binary grids; -20 to +5 dB in 1 dB steps
Dataset	104,000 images: 4,000 per SNR
Identical data split	62,400 training / 20,800 validation / 20,800 test
Training	8 epochs; batch size 64; Adam; learning rate 0.001; seed 42
Model selection	Minimum validation loss; no early stopping or learning-rate schedule

The data generator was unchanged. No model receives SNR as an input. U-Net, EvidenceAxialNet and FastUNet use BCE + Dice loss; FCNN uses MSE on bounding-box coordinates.

Model configurations

Model	Training configuration
Original U-Net	Baseline U-Net implementation; native BatchNorm backend
EvidenceAxialNet	Base width 16; evidence features enabled; 3 refinement steps
FastUNet	Base width 8; multi-branch training blocks; native BatchNorm backend
FCNN recreation	5,000 inputs; four 256-unit hidden layers; 80 outputs (20 boxes); dropout 0.4

Use the saved model-source snapshots, configurations, dataset manifests and split indices from the original runs to reproduce these settings. The same seed alone does not guarantee identical results on different hardware.